

Hackathon Brief — Social Immersive Digital Twin of Shuk HaTikvah

Goal

Over three days, develop a working prototype of an **immersive social digital twin of the market** that connects the 360° field recording with spatial/GIS information and AI-derived social, material and operational observations. The prototype should demonstrate that the market can be explored not only as physical space, but as a changing **social–material–spatial system**.

The existing demo provides a strong base: 360° navigation, person detection, shop tags, material segmentation and GIS positioning. The hackathon should concentrate on making these layers work together inside the immersive environment rather than as separate analytical panels.

Core deliverables

1. One integrated immersive viewer. Information should appear **inside the market view itself**, spatially anchored to people, shops, objects and locations. The user should be able to rotate and move through the scene without losing the analytical information. Ideally, the same viewpoint can transition between **360° photographic/video view → simplified/digital representation → fully illustrated interpretation**, while retaining the same shop tags, people/activity markers and analytical overlays.

2. Persistent spatial identification. Shops, people, activities and significant objects should be tied to a consistent location/ID so they can later be exported and analysed in GIS. At minimum: shop ID/type, position along route, timestamp/frame, and confidence. This is essential for subsequently creating market morphologies rather than analysing isolated images.

3. Three analytical layers.
Social/human: people count and density; approximate broad age groups where visually defensible; vendor/shopper/passer-by/worker; visible interactions and their type; gathering/stopping points.
Material: shop/stall morphology as well as trade type — walk-in shop, open stall, counter, temporary table, etc.; vendor inside/outside; shop-front permeability; signage; construction/display materials; and importantly **waste materials** such as cardboard, plastic, organic waste, crates and packaging.
Spatial: circulation, thresholds, junctions, shop spill-out, obstacles, seating, active frontages and areas of concentrated interaction.

4. Operational/logistics activity detection. Identify and mark market functions that are easily missed if the analysis focuses only on customers and vendors: **delivery workers, porters, box/crate collectors, road sweepers/cleaners, waste handlers, loading/unloading, stock movement, e-bike deliveries and other service activity**. These should be spatially and temporally tagged so they can later be mapped as part of the market's operational metabolism.

5. Interaction with the information. Clicking or selecting an identified shop/person/object should reveal a concise information card within the immersive view rather than forcing the user to look away to a separate dashboard. The interface should allow analytical layers to be switched on/off independently: **people/interactions, shops, materials/waste, logistics/operations, GIS tags, researcher drawings**.

Key research test

The prototype does **not** need to solve full cultural interpretation. Its purpose is to demonstrate that AI/VLM analysis can turn 360° footage into structured, spatially anchored observations that can later be compared in GIS.

The most interesting outcome would be the ability to begin asking questions such as:

Do open stalls produce more social interaction than walk-in shops? Do vendors positioned outside interact more frequently? Are older and younger users concentrated in different parts of the market? Do interaction hotspots correspond with particular spatial forms, signage, materials or shop types? Where do waste, logistics and social activity overlap?

Success at the end of Day 3

A successful hackathon outcome is a **short navigable market segment** in which a user can rotate through the 360° environment, switch between photographic/digital/illustrated representations, see persistent spatially anchored analytical information, and export or connect those observations to GIS.

The key proof of concept is therefore:

360° field recording → AI/VLM observation → spatial anchoring → immersive visualisation → GIS-ready social/material/spatial data.